EDS 223: Homework Assignment 3

United States West Coast Aquaculture Analysis

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Introduction

This assignment investigates which Exclusive Economic Zones (EEZ) on the West Coast of the United States have suitable sea surface temperature (SST) and depth values for developing marine aquaculture for multiple species of oysters and Yesso Scallop (*Mizuhopecten yessoensis*).

Load Packages

```
library(sf)          # vector handling
library(terra)       # raster handling
library(tidyverse)
library(tmap)        # map making
```

Prepare Data

Load Data

```
# Load sea surface temperature data for 2008 to 2012
sst_2008 <- terra::rast('data/average_annual_sst_2008.tif')
sst_2009 <- terra::rast('data/average_annual_sst_2009.tif')
sst_2010 <- terra::rast('data/average_annual_sst_2010.tif')
sst_2011 <- terra::rast('data/average_annual_sst_2011.tif')
sst_2012 <- terra::rast('data/average_annual_sst_2012.tif')

# Stack sea surface temperature rasters
sst_2008_to_2012 <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)

# Load ocean depth data
ocean_depth <- terra::rast('data/depth.tif')
```

```
# Load Exclusive Economic Zones data
eez <- sf::st_read('data/wc_regions_clean.shp')
```

Check Coordinate Reference Systems

```
# Check if sea surface temperature and ocean depth raster CRSs match

if(st_crs(sst_2008_to_2012) == st_crs(ocean_depth)) {
  print('CRS match')
}else{
  warning('CRS do not match')
  print(paste('sst_2008_to_2012 CRS is', st_crs(sst_2008_to_2012)))
  print(paste('ocean_depth CRS is', st_crs(ocean_depth)))
}
```

```
[1] "CRS match"
```

```
# Check if Exclusive Economic Zone data and sea surface temperature data CRSs
match
if(st_crs(eez) == st_crs(sst_2008_to_2012)) {
  print('CRS match')
}else{
  warning('CRS do not match')
  print(paste('eez CRS is', st_crs(eez)))
  print(paste('sst_2008_to_2012 CRS is', st_crs(sst_2008_to_2012)))
}
```

```
[1] "CRS match"
```

Process Data

Create raster of average sea surface temperature

```
# Create single raster of average sea surface temperature
sst_mean <- terra::app(sst_2008_to_2012, fun = mean, na.rm = TRUE)

# Convert mean sea surface temperature to degrees C
sst_mean_celcius <- sst_mean - 273.15
```

Crop ocean depth raster to match extent of sea surface temperature raster

```
ocean_depth <- crop(ocean_depth, sst_mean_celcius)
```

Match ocean depth raster resolution to resolution of sea surface temperature raster

```
# Use bilinear method to resample ocean_depth and match sst_mean_celcius resolution
ocean_depth_resampled <- resample(ocean_depth, sst_mean_celcius, method = 'bilinear')

# Check that ocean depth and sea surface surface temperature rasters match in resolution, extent, and coordinate reference system
cat("After cropping and resampling ocean_depth_resampled:\n")
```

```
After cropping and resampling ocean_depth_resampled:
```

```
cat("Resolutions match:", all(res(ocean_depth_resampled) == res(sst_mean_celcius)), "\n")
```

```
Resolutions match: TRUE
```

```
cat("Extents match: ", ext(ocean_depth_resampled) == ext(sst_mean_celcius), "\n")
```

```
Extents match:  TRUE
```

```
cat("Coordinate reference systems match: ", crs(ocean_depth_resampled) == crs(sst_mean_celcius))
```

```
Coordinate reference systems match:  TRUE
```

Create a function to generate a map displaying amount of West Coast EEZ area suitable for aquaculture

```
map_suitable_areas <- function(temp_min, temp_max, depth_min, depth_max, species){

  # Define reclassification matrix for sea surface temperature
  sst_rcl <- matrix(c(-Inf, temp_min, 0,
                      temp_min, temp_max, 1,
                      temp_max, Inf, 0),
                    ncol = 3, byrow = TRUE)
```

```r
  # Define reclassification matrix for ocean depth
  depth_rcl <- matrix(c(-Inf, depth_min, 0,
                depth_min, depth_max, 1,
                depth_max, Inf, 0),
                ncol = 3, byrow = TRUE)

  # Classify areas where sea surface temperature is suitable for species
  sst_suitable <- classify(sst_mean_celcius, rcl = sst_rcl)

  # Classify areas where depth is suitable for species
  depth_suitable <- classify(ocean_depth_resampled, rcl = depth_rcl)

  # Stack rasters
  suitable_sites <- c(sst_suitable, depth_suitable)

   # Create new raster where sea surface temperature and depth are suitable for species
  suitable_sites <- lapp(suitable_sites, fun = function(s, d) (s*d))

  # Rasterize Exclusive Economic Zone data and set region ID as cell values
  eez_raster <- rasterize(eez, suitable_sites, field = "rgn_id")

  # Calculate area of suitable sites for aquaculture in square kilometers
  suitable_sites_area <- cellSize(suitable_sites, unit = 'km', mask = TRUE)

  # Mask cells in suitable_sites_area based on suitable_sites, change value of masked cells to 0
   suitable_sites_area <- mask(suitable_sites_area, suitable_sites, maskvalues = 0)

  # Sum area of suitable sites by eez region
   suitable_area_by_eez <- zonal(suitable_sites_area, eez_raster, fun = 'sum', na.rm = TRUE)

  # Join suitable_area_by_ez with original eez data
  suitable_area_by_eez <- tibble(suitable_area_by_eez)

  suitable_area_by_eez <- inner_join(suitable_area_by_eez, eez, by = 'rgn_id') |>
    mutate(suitable_area_km = area) |> # Assign area column values to suitable_area_km
  select(-c(area, rgn_key, area_m2, area_km2)) |> # Remove unwanted columns
```

```
  st_as_sf() # Convert to simple feature to recognize geometries

  suitable_area_map <- tm_basemap("OpenTopoMap") +
    tm_shape(suitable_area_by_eez) +
    tm_options(component.autoscale = FALSE) +
    # Map exclusive economic zones by suitable area
    tm_polygons(fill = 'suitable_area_km', lwd = 0.5, fill_alpha = 0.8,
              fill.scale = tm_scale_categorical(
             values = c('#edf8fb', '#b2e2e2', '#66c2a4', '#2ca25f', '#006d2c')),
              fill.legend = tm_legend(title = "Suitable Area,\nSquare km",
                                      title.size = 0.7,
                                      position = tm_pos_out('right'))) +
    tm_compass(position = c('left', 'bottom'), # Add compass
                size = 2) +
    tm_scalebar(position = c('left', 'bottom')) + # Add scalebar
   tm_title(text = paste('Area of West Coast Exclusive Economic Zones\nSuitable
for', species, 'Aquaculture'),
              size = 1,
              fontface = 'bold',
              position = tm_pos_out('center', 'top'))

  suitable_area_map
}
```

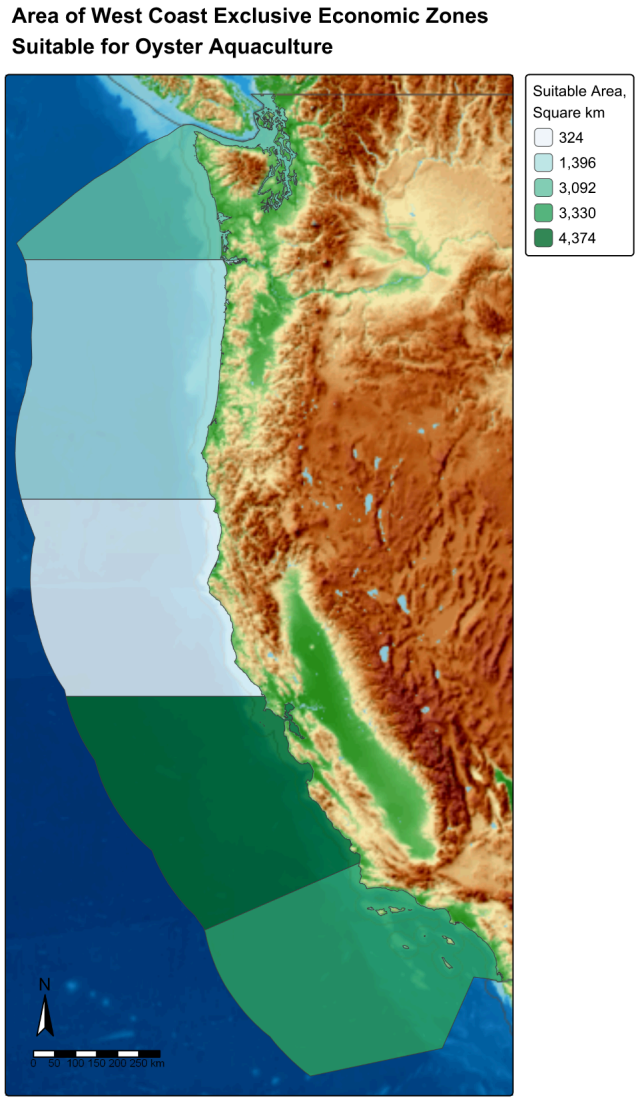
Map suitable locations for marine aquaculture for species of interest

Oysters

Oysters require the following conditions for optimal growth:

- Sea surface temperature in the range of 11 to 30 degrees Celsius
- Depth of 0 to 70 meters below sea level

```
# Map areas suitable for oyster aquaculture
map_suitable_areas(temp_min = 11, temp_max = 30, depth_min = -70, depth_max =
0, species = 'Oyster')
```



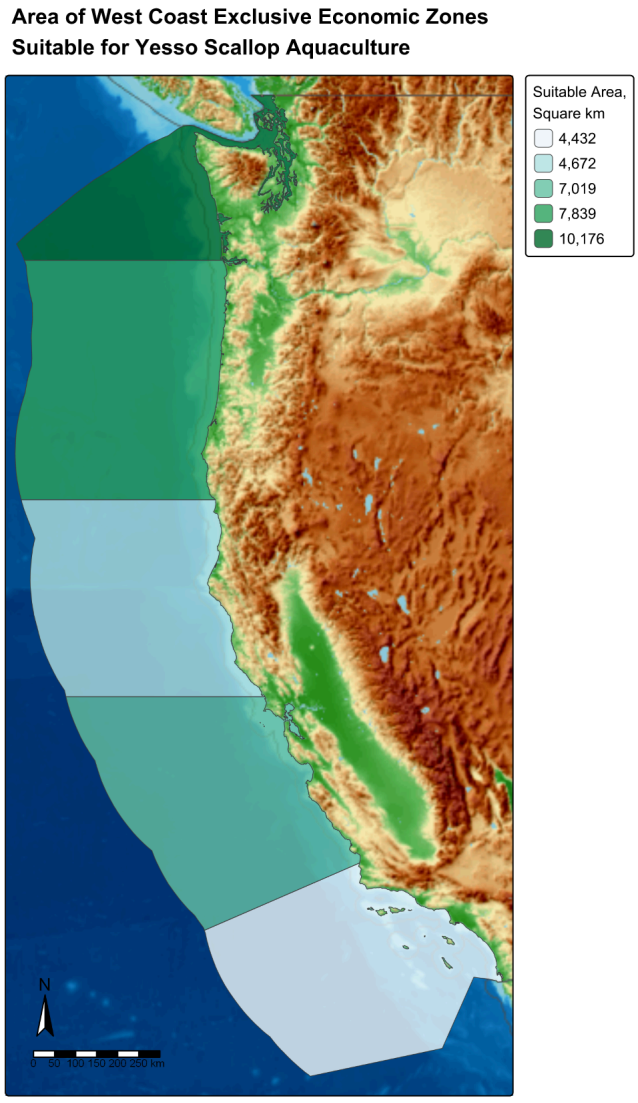
While all west coast Exclusive Economic Zones have areas that are suitable for oyster aquaculture, the areas off the central and southern coast of California contain the greatest amount of suitable area. 4,374 square kilometers of suitable area were identified in the Central California Exclusive Economic Zone and 3,330 square kilometers of suitable area were identified in the Southern California Exclusive Economic Zone.

Yesso scallops

Yesso scallops (*Mizuhopecten yessoensis*) require the following conditions for optimal growth:

- Sea surface temperature in the range of 7 to 17 degrees Celsius
- Depth of 0 to 100 meters below sea level

```
# Map areas suitable for Yesso scallop aquaculture
map_suitable_areas(temp_min = 7, temp_max = 17, depth_min = -100, depth_max =
-0, species = 'Yesso Scallop')
```



Each of the west coast Exclusive Economic Zones contain at least 4,000 square kilometers of area suitable for Yesso scallop aquaculture. The Washington Exclusive Economic Zone ranked the highest, with a total of 10,176 square kilometers of area suitable for Yesso scallop aquaculture identified.

Data Sources

Sea surface temperature data from NOAA's Daily Global 5km Satellite Sea Surface Temperature Anomalyv3.1, bathymetry data from General Bathymetric Chart of the Oceans, and Exclusive Economic Zone data from Marineregions.org were used in this assignment.

Oyster conditions for optimal growth were sourced from the EDS 223 Homework 3 Assignment page.

Yesso scallop conditions for optimal growth were sourced from Food and Aquaculture Organization of the United Nations (temperature values) and SeaLifeBase (depth values).